

NEBULA and NEBULA-Plus geometry, performance, and data-source notes

(collected for the `kondo_smsimulator` / `smsimulator5.5` project)

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2026-05-15

1 Layout and orientation (executive summary)

Along the beam direction, from target downstream:

target $\rightarrow$ **NEBULA-Plus (front)** $\rightarrow$ **NEBULA (back)**.

That is, **NEBULA-Plus sits in front of NEBULA** (between FDC2/HOD and NEBULA). NEBULA-Plus contains **90 plastic-scintillator neutron bars**, i.e. **30 bars fewer** than NEBULA's 120 bars. The plastic bar dimensions (120 mm $\times$ 1800 mm $\times$ 120 mm) and the veto layout (10 mm plastic scintillator at the upstream face of each wall) are otherwise identical to NEBULA.

NEBULA-Plus was built by a Japan / LPC-Caen collaboration under the French ANR EXPAND project, shipped to the RIBF in summer 2022 and mechanically installed in October 2022. Initial in-beam commissioning with $^7\text{Li}(p,n)^7\text{Be}$ was scheduled for the next NEBULA campaign [2].

NEBULA-Plus is now **implemented in smsimulator5.5** (branch `nebula-plus-impl` merged to `main` on 2026-05-16, tags `nebula-plus-phase1/2/3`). End-to-end verification ran 100 200 MeV pencil-beam neutrons through both detectors and recovered neutron candidates from three reco classes (NEBULA-only, NEBULA-Plus-only, joint); the joint reco produced $\sim \times 1.86$ as many candidates as NEBULA alone, consistent with the qualitative “ $\sim \times 2$ ” gain quoted in Tanaka 2024 [3]. See §6 and §7 for full implementation details and the source of every authoritative parameter.

2 Internal geometry

2.1 NEBULA (existing, already in smsimulator5.5)

Two walls (Wall-A, Wall-B), each with two sub-layers of neutron bars and one upstream veto layer:
Total: 120 Neut bars + 24 Veto bars = 144 counters.

2.2 NEBULA-Plus (now implemented in smsimulator5.5)

From the LPC-Caen / RIKEN nptool plugin `commissioning/detector.yaml` (see §5), the four sub-layers of NEBULA-Plus are:

Total: 22 + 22 + 23 + 23 = 90 Neut bars + 24 Veto bars = **114** counters, i.e. **30 Neut bars fewer than NEBULA**.

2.3 Absolute Z of the NEBULA-Plus origin

The `detector.yaml` files in the LPC-Caen nptool plugin use a placeholder target block ($x = 1$, $y = 2$, $z = 3$ mm); the absolute Z of the NEBULA-Plus origin relative to the SAMURAI target is therefore *not* captured in any of the config files we have read-access to. The home directories of the responsible authors (`/home/kondo`, `/home/otsu`, `/home/gibelin`, `/home/jlemarie`) on `ribfana04` were permission-denied.

Table 1: NEBULA layout (s021 configuration; coordinates from the smsimulator5.5 geometry/s021/NEBULA_Detectors_samurai21.csv).

Element	# bars	Z rel. NEBULA origin (mm)	X range (mm)
Wall-A Veto	12	−275, −260	—
Wall-A Sub1	30	0	+1908 → −1641.6
Wall-A Sub2	30	+130	same
Wall-B Veto	12	+575.3, +590.3	—
Wall-B Sub1	30	+850.3	same
Wall-B Sub2	30	+980.3	same
NEBULA origin in target frame (s021)	$Z = 9784$ mm		
Bar size (Neut)	$120 \times 1800 \times 120$ mm ($W \times H \times L$)		
Bar pitch in X	122.4 mm (i.e. 2.4 mm air gap)		
Bar centroid Y	−35.4 mm (slightly below beam axis; see NEBULA_Pos_FromMagCenter)		
Veto size	$320 \times 1900 \times 10$ mm		
Time resolution per PMT	$\sigma_t = 0.240$ ns (NEBULA_samurai21.csv)		
Bar material (in current sim)	G4_PLASTIC_SC_VINYLTOLUENE (NIST PVT analog of BC408/EJ200)		

Table 2: NEBULA-Plus layout from the nptool plugin’s detector.yaml. Z is given in the NEBULA-Plus-internal frame (its own Wall-A Sub1 is the origin). X_0 is the centre of the leftmost bar; the rest of the row follows $X_i = X_0 + i \times 120$ mm.

Wall	SubLayer	# bars	X_0 (mm)	X range (mm)	Z (mm)
A	Sub1	22	−1320	−1320 → +1200	0
A	Sub2	22	−1320	−1320 → +1200	130
B	Sub1	23	−1320	−1320 → +1320	845
B	Sub2	23	−1320	−1320 → +1320	975
Bar size (Neut): $120 \times 1800 \times 120$ mm — identical to NEBULA					
Bar material: BC400 in nptool (essentially equivalent to NEBULA’s PVT analog)					
Bar pitch in X : 120 mm (no air gap in the plugin model; InterModule = 1 mm declared but unused)					
Veto size: $320 \times 1900 \times 10$ mm					
Veto distance to wall: 100 mm upstream of the corresponding Neut wall (WallToVeto = 10 cm)					
Veto count: 12/wall (the plugin selects 12 when $N_{\text{bars}} > 15$, else 6)					
PMT: Hamamatsu H11284 assemblies; FASTER digital electronics developed at LPC-Caen					

Chosen value used in smsimulator5.5 (placeholder; survey-grade value still to be obtained from LPC-Caen):

$$Z_{\text{NEBULA-Plus, origin}} = 8089 \text{ mm} \quad (\text{target frame}),$$

selected so the Wall-A–Sub1 spacings $845 \rightarrow 850 \rightarrow 850.3$ mm walk uniformly from NEBULA-Plus into NEBULA:

The chosen 850 mm NPL-B→NEB-A gap can be overridden at runtime via /samurai/geometry/NEBULAPlus/P 0 0 <Z> mm (and similarly for NEB) once a real survey is provided.

3 Material / electronics summary

- Plastic: BC400 (NEBULA-Plus) and BC408/EJ200 (NEBULA) are PVT-based scintillators; densities and light yields agree to better than 1%. In the Geant4 simulation both can be represented by G4_PLASTIC_SC_VINYLTOLUENE.

Table 3: NEBULA-Plus+NEBULA target-frame layout used in smsimulator5.5 (combined of §2.2 numbers and chosen Z_0).

Wall / sub-layer	Z (mm)	Sub1→Sub1 spacing (mm)
NPL Wall-A Veto	7989	—
NPL Wall-A Sub1	8089	—
NPL Wall-A Sub2	8219	—
NPL Wall-B Veto	8834	—
NPL Wall-B Sub1	8934	845.0 (NPL internal)
NPL Wall-B Sub2	9064	—
NEB Wall-A Veto	9509	—
NEB Wall-A Sub1	9784	850.0 (gap chosen)
NEB Wall-A Sub2	9914	—
NEB Wall-B Veto	10359.3	—
NEB Wall-B Sub1	10634.3	850.3 (NEB internal)
NEB Wall-B Sub2	10764.3	—

- Refractive index $n = 1.58$; bulk attenuation length 6680 mm (declared in `NebulaPlusGeant4.cxx`).
- PMTs: NEBULA-Plus uses Hamamatsu H11284; NEBULA uses Hamamatsu H7195 or equivalent (depending on bar batch).
- DAQ: NEBULA-Plus uses the FASTER digital readout developed at LPC-Caen, integrated with SAMURAI via a coincidence trigger derived from the two-PMT signal pair of each bar [2].

4 Detection efficiency

4.1 NEBULA alone

Single-neutron detection efficiency at $E_n = 200$ MeV:

$$\varepsilon_{1n}^{NEBULA} = 32.5 \pm 0.3 \text{ (stat)} \pm 0.9 \text{ (syst)} \%,$$

measured with the ${}^7\text{Li}(p,n){}^7\text{Be}$ reaction (Kondo et al., NIM B 463 (2020) 173–178 [1]).

4.2 NEBULA $\cup$ NEBULA-Plus (combined)

The NEBULA+NEBULA-Plus array effectively doubles the number of layers (four neutron sub-layers $\rightarrow$ eight). The Tanaka et al. review (arXiv:2412.17887 [3]) states, qualitatively:

- **1-neutron** efficiency: increases by **nearly a factor of 2** (i.e. from $\sim 32.5\%$ to $\sim 60\text{--}65\%$ at 200 MeV, rough extrapolation).
- **4-neutron** efficiency: improves by **almost one order of magnitude** relative to NEBULA alone.

Absolute combined numbers and the energy dependence have to come from the in-beam commissioning data (NEBULA-Plus alone ε_{1n} to be determined once the ${}^7\text{Li}(p,n)$ run is analysed — planned but not yet published as of the APR 2022 report).

5 Local file paths (where this information lives)

5.1 On this workstation

smsimulator5.5 NEBULA G4 module `kondo_smsimulator/sources/smg4lib/detectors/NEBULA/`

NEBULA geometry CSVs (s021) kondo_smsimulator/work/geometry/s021/NEBULA_samurai21.csv
kondo_smsimulator/work/geometry/s021/NEBULA_Detectors_samurai21.csv

anaroot NEBULA parameter db anaroot/db/NEBULA.20250625.xml (and dated siblings)
anaroot/db/NEBULA_Pos_FromMagCenter.csv
anaroot/db/NEBULA.ods

anaroot NEBULA-Plus placeholder (only veto names V301–V410 at $Z = 0$) anaroot/db/NEBULA.20250625.xml
anaroot/src/sources/Reconstruction/SAMURAI/src/TArtSAMURAIParameters.cc:544
(comment: // for new config. with NEBULA Plus)

Papers (downloaded) docs/papers/NEBULA-Plus_Kondo_RIKEN_APR2022.pdf
docs/papers/NEBULA-Plus_review_Tanaka2024_arxiv2412.17887.pdf
docs/papers/multi-neutron_SAMURAI_Yang2019_arxiv1903.11256.pdf
docs/papers/NEBULA_RIBF_factsheet.pdf
/docs/papers/NEBULA_indico_talk.pdf

5.2 On the analysis server ribfana04

These are the definitive NEBULA-Plus geometry and code locations on the s053 experiment account (read-access only):

nptool plugin (Geant4 + Physics) /home/s053/exp/exp2301_s053/nptool/plugin/nebula-plus/sources/Geant4Physics
Main files:

- NebulaPlusGeant4.cxx, NebulaPlusGeant4.h — Geant4 module / veto construction, materials, vis attributes.
- NebulaPlusDetector.cxx, NebulaPlusDetector.h — yaml parsing, AddLayer / AddSBT / AddSBV; Z is relative to target.
- NebulaPlusPhysics.h, NebulaPlusData.h/cxx, NebulaPlusSpectra.cxx, NebulaPluslinkdef.hh.

detector configurations /home/s053/exp/exp2301_s053/nptool/commissioning/detector.yaml
(the only one with X_start = -1320mm; s053/detector.yaml and sandbox/detector.yaml use X = 0.)

Calibrations /home/s053/exp/exp2301_s053/nptool/commissioning/calibration/
NebulaPlus_Energy.cal, NebulaPlus_Up_Energy.cal, NebulaPlus_Down_Energy.cal,
NebulaPlus_Tdiff_Offset.cal, NebulaPlus_Tdiff_Offset_Jan2023.cal

Online monitor (s053) /home/s053/exp/exp2301_s053/sm_online_utils/s053/chknebulaplus/OnlineMonitor
(counts/rates only; no geometry constants).

Raw data + FASTER setup files /home/s053/rawdata/nebulaplus/NEBULA_Test_2025*.fast
(+.setup)

FASTER electronics source tree /home/s053/exp/exp2301_s053/faster/faster/fasterac-2.18/

Rsync from NEBULA-Plus DAQ host /home/s053/exp/exp2301_s053/scripts/rsync_expand_test.sh
pulls from expand@expandacq: ./Projects/test_veto/data/Test_VETO/ into \$HOME/rawdata/nebulaplus

5.3 External

- Project page (LPC-Caen): <https://nebula-plus.in2p3.fr/> (with /manual/ and /documentation-and-procedures/ sub-areas behind login).
- RIKEN APR 56 (2023) p. 91 – installation report: <https://www.nishina.riken.jp/researcher/APR/APR056/pdf/91.pdf>

6 Implementation in smsimulator5.5

Implemented on branch `nebula-plus-impl` (merged to main 2026-05-16, tags `nebula-plus-phase1/2/3` mark each phase boundary). The corresponding spec and plan are `docs/superpowers/specs/2026-05-15-nebula` and `docs/superpowers/plans/2026-05-15-nebula-plus.md`.

6.1 Build environment

```
# 1. Activate the anaroot conda environment (Geant4 11.x, ROOT 6.36,
# Xerces-C, anaroot -- all linked to that env)
micromamba activate anaroot-env

# 2. spdlog must be installed in the env (used by smlogger). One-time:
micromamba install -n anaroot-env -c conda-forge spdlog

# 3. Source project setup (defines SMSIMDIR, TARTSYS, paths)
cd $HOME/workspace/dpol/smsimulator5.5
source setup.sh

# 4. Configure + build
mkdir -p build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release \
        -DBUILD_TESTS=ON -DBUILD_APPS=ON \
        -DBUILD_ANALYSIS=ON -DWITH_ANAROOT=ON
cmake --build . -j$(nproc)
```

Important: TARTSYS must be set *before* `cmake ..` runs; otherwise WITH_ANAROOT silently flips off and the NEBULASimDataConverter_TArtNEBULAPla translation unit fails to find TArtNEBULAPla.hh. `setup.sh` on host TBTdebian exports TARTSYS=/home/tian/software/anaroot/install.

6.2 Geometry configuration (CSV)

Two pairs of CSV files drive the geometry. The smsim G4 messengers read them at runtime via `/samurai/geometry/.../ParameterFileName`.

Table 4: Geometry-CSV inventory under `smsimulator5.5/configs/simulation/geometry/`. Each pair is {global-parameters file, per-bar table}.

Detector	Global params	Per-bar table
NEBULA (Dayone)	NEBULA_Dayone.csv	NEBULA_Detectors_Dayone.csv
NEBULA-Plus (s021)	s021/NEBULAPlus_samurai21.csv	s021/NEBULAPlus_Detectors_samurai21.csv

The NEBULA-Plus CSV pair was generated from the numbers in §2.2. The global-params file content is shown below verbatim; the per-bar file has 114 rows (90+24) following the `Layer/SubLayer/PositionX/` schema also used by NEBULA.

```
% configs/simulation/geometry/s021/NEBULAPlus_samurai21.csv
Position, 0, 0, 8089,
NeutSize, 120, 1800, 120,
VetoSize, 320, 1900, 10,
Angle, 0, 0, 0,
////input time reso for one PMT
TimeReso, 0.240,
```

6.3 Geant4 module classes (libs/smg4lib/)

NEBULA-Plus’s G4 module mirrors NEBULA’s file-for-file. The construction-side classes live in `src/construction/{src,include}/`; the data-side classes live in `src/data/{src,include}/` with mirror public headers also in `libs/smg4lib/include/`.

Table 5: G4-side class inventory. NEBULA-Plus files were created by sed-clone of the corresponding NEBULA files in branch `nebula-plus-impl` Phase 1.

Role	NEBULA class	NEBULA-Plus class
G4 geometry	NEBULAConstruction	NEBULAPlusConstruction
G4 UI messenger	NEBULAConstructionMessenger	NEBULAPlusConstructionMessenger
Sensitive detector	NEBULASD	NEBULAPlusSD
CSV reader	NEBULASimParameterReader	NEBULAPlusSimParameterReader
SimData branch reg.	NEBULASimDataInitializer	NEBULAPlusSimDataInitializer
Step→hit conv.	NEBULASimDataConverter_TArtNEBULAPla	NEBULAPlusSimDataConverter_TArtNEBU
Run-param ROOT cls	TNEBULASimParameter	TNEBULAPlusSimParameter
Hit ROOT cls	TArtNEBULAPla (anaroot)	TArtNEBULAPlusPla (smsim, new)

Both detectors are instantiated and attached to the SAMURAI experimental hall in `libs/sim_deuteron_core/` (NPL added at Phase 1 task P1.11). The corresponding SimData initializer and converter are registered in `apps/sim_deuteron/main.cc` (P1.12).

6.4 Runtime messenger commands

```
# Load both detectors (works in any .mac executed by bin/sim_deuteron)
/samurai/geometry/NEBULA/ParameterFileName {SMSIMDIR}/configs/simulation/geometry/NEBULA_Dayone.csv
/samurai/geometry/NEBULA/DetectorParameterFileName {SMSIMDIR}/configs/simulation/geometry/
  NEBULA_Detectors_Dayone.csv

/samurai/geometry/NEBULAPlus/ParameterFileName {SMSIMDIR}/configs/simulation/geometry/s021/
  NEBULAPlus_samurai21.csv
/samurai/geometry/NEBULAPlus/DetectorParameterFileName {SMSIMDIR}/configs/simulation/geometry/s021/
  NEBULAPlus_Detectors_samurai21.csv

# Optionally override the global Z when a real LPC survey lands
/samurai/geometry/NEBULAPlus/Position 0 0 8089 mm

/samurai/geometry/Update
```

A minimal smoke macro is shipped at `configs/simulation/macros/test_nebula_plus.mac`.

6.5 Output ROOT-tree branches

Each event of `bin/sim_deuteron` produces (relevant branches only):

NEBULAPla TClonesArray of TArtNEBULAPla (anaroot class) — NEBULA hits, with `Layer` = 3 (wall A) or 4 (wall B).

NEBULAPlusPla TClonesArray of TArtNEBULAPlusPla (smsim class) — NEBULA-Plus hits, with `Layer` = 1 (wall A) or 2 (wall B).

TNEBULASimParameter, TNEBULAPlusSimParameter per-run snapshots of the detector parameter map.

6.6 Reconstruction (`libs/analysis/`)

The legacy `NEBULAREconstructor` was refactored in Phase 2 into a small hierarchy:

NEBULABaseReco abstract base. Holds public algorithm steps shared by all derived classes: `ClusterHits`, `ReconstructFromCluster`, `CalculateBeta`, `CalculateNeutronEnergy`, `ApplyPositionSmearing`, `ApplyTimeSmearing`. Pure virtual: `ExtractHits()`.

NEBULAReco single-detector NEBULA; ExtractHits reads TClonesArray<TArtNEBULAPla>. Preserves the legacy behaviour bit-for-bit (regression-pinned by test_NEBULAReco_legacy_compat).

NEBULAPlusReco single-detector NEBULA-Plus; ExtractHits reads TClonesArray<TArtNEBULAPlusPla>, tags wall_tag 1/2.

NebulaJointReco takes both detector inputs and merges them through a single ExtractHits call; clusterer sees all 8 sub-layers and tags wall_tag 1-4.

Default parameter values (preserved from the legacy class): fTimeWindow=10.0 ns, fEnergyThreshold=1.0 MeV, fPositionSmearing=5.0 mm, fTimeSmearing=0.5 ns.

The reco *algorithms* are now independent of anaroot’s TArtRecoNeutron / TArtCalibNEBULA. Anaroot is still a build-time dependency because the NEBULA-hit data class TArtNEBULAPla lives in anaroot and is used as the on-disk schema for the NEBULAPla branch. Removing that dependency is left as a follow-up (would require introducing a smsim-side mirror class for NEBULA, plus a schema-evolution step for existing .root files).

6.7 End-to-end verification (2026-05-16)

Ran 100 monoenergetic 200 MeV neutrons through the NEBULA-Plus+NEBULA geometry; see tests/integration/test_sim_deuteron_with_nebula_plus.sh.

Table 6: End-to-end sim+reco numbers (100 incident 200 MeV neutrons, pencil beam from $Z = -2000$ mm).

Stage / class	Events with hits/candidates	Total entries
NEBULA hits in tree	51	159 hits
NEBULA-Plus hits in tree	55	227 hits
NEBULAReco candidates	32	35
NEBULAPlusReco candidates	39	42
NebulaJointReco candidates	55	65

The joint-to-NEBULA candidate ratio is $65/35 \approx 1.86$, consistent with the qualitative “ $\sim \times 2$ ” single-neutron gain quoted in the Tanaka 2024 review [3].

6.8 Tests

Table 7: New tests added by Phases 1–3 (ctest labels in parentheses).

File	Cases
tests/unit/test_NEBULAPlusSimParameterReader.cc (unit)	3
tests/unit/test_NEBULAPlusReco_basic.cc (unit)	3
tests/unit/test_NebulaJointReco_combined.cc (unit)	3
tests/analysis/test_NEBULAReco_legacy_compat.cc (analysis)	1
tests/integration/test_sim_deuteron_with_nebula_plus.sh (integration)	1

After Phases 1–3, the project ctest count went from **18 passing / 20 failing** to **30 passing / 20 failing** (+12 passing, 0 regression on previously-passing tests). The 20 pre-existing failures (ParticleTrajectoryTest.*, TargetReconstructorTest.*) are unrelated to NEBULA-Plus and remain out-of-scope here.

7 Provenance: where each NPL number came from

For NEBULA-specific numbers (Z=9784 mm origin, 122.4 mm X pitch, 35.4 mm Y offset, 0.240 ns time resolution), the authoritative source is the existing CSV `configs/simulation/geometry/NEBULA_Dayone.csv` and the anaroot db files listed in §5; the design paper is Kondo 2020 NIM B [1].

References

- [1] Y. Kondo, T. Tomai, T. Nakamura, “Recent progress and developments for experimental studies with the SAMURAI spectrometer,” Nucl. Instrum. Methods Phys. Res. B **463** (2020) 173–178. DOI: 10.1016/j.nimb.2019.04.085.
- [2] N. A. Orr, J. Gibelin, Y. Kondo, H. Otsu, on behalf of the NEBULA-Plus collaboration, “Installation of the NEBULA-Plus neutron array,” RIKEN Accel. Prog. Rep. **56** (2023) p. 91.
- [3] J. Tanaka et al., “Detectors for next-generation quasi-free scattering experiments at the RIBF,” arXiv:2412.17887 (to appear in PTEP).
- [4] T. Nakamura, Y. Kondo, “Large acceptance spectrometers for invariant mass spectroscopy of exotic nuclei and future developments,” Nucl. Instrum. Methods Phys. Res. B **376** (2016) 156. DOI: 10.1016/j.nimb.2016.01.003.

Table 8: Information sources for every authoritative NEBULA-Plus parameter used in smsimulator5.5.

Parameter	Value	Source
Bar dimensions ($W \times H \times L$)	$120 \times 1800 \times 120$ mm	NebulaPlusGeant4.cxx
Veto dimensions	$320 \times 1900 \times 10$ mm	NebulaPlusGeant4.cxx
Number of Neut bars per sub-layer	22, 22, 23, 23	commissioning/detector/plugin)
Number of Veto bars per wall	12	NebulaPlusGeant4.cxx (m_NbrModule>15⇒12)
Bar pitch in X	120 mm (touching)	NebulaPlusDetector: by m_BarWidth = 120
Veto pitch in X	310 mm	NEBULA convention Veto type)
Veto distance to wall	100 mm upstream of Neut Sub1	NebulaPlusGeant4.cxx
Sub-layer Z gap	130 mm	commissioning/detector/tries at Z=0/130/845/
X -start of first bar	−1320 mm (asymmetric)	commissioning/detector/-132 0 0 cm)
Bar material (G4)	G4_PLASTIC_SC_VINYLTOLUENE	Same as NEBULA in sufficient PVT analog
Refractive index n	1.58	NebulaPlusGeant4.cxx
Bulk attenuation length	6680 mm	NebulaPlusGeant4.cxx
PMT time resolution per side	0.240 ns	NEBULA value reused commissioning number yet
PMT model	Hamamatsu H11284	APR 56 (2023) p. 91 [2]
Electronics	FASTER (LPC-Caen)	APR 56 (2023) p. 91 [2]
Absolute Z of NPL origin (target frame)	8089 mm	Chosen for smsim Sub1→Sub1 spacing mm across NPL-A-A→NEB-B; survey-graph be obtained from LPC anaroot db convention V401-V410 pre-reserved TArtSAMURAIParameters
Layer index assignment	NPL: L=1,2; NEB: L=3,4	Chosen for smsim5.5 (L index; Veto distinguishing prefix and SubLayer=0
Bar/Veto ID ranges	101–122, 201–222, 301–323, 401–423; 501–512, 601–612	Tanaka 2024 [3] (at PTEP, to appear). Confirmed by the smsim end-to-end Tanaka 2024 [3]; not in smsim (requires a breakup macro)
Single-neutron efficiency gain	$\sim \times 2$ vs NEBULA alone	APR 56 (2023) p. 91 [2]
Four-neutron efficiency gain	$\sim \times 10$ vs NEBULA alone	
NPL installation date	October 2022	